

in the domain such that $P(c) \wedge \forall y (\neg P(y) \rightarrow c=y)$ is True.

By defn. of universal quantifier, for any arbitrary element c_2 in the domain,

$P(c) \wedge (P(c_2) \rightarrow c=c_2)$ is True. By defn. of AND, $P(c)$ is True and $(P(c_2) \rightarrow c=c_2)$ is True for any arbitrary c_2 and for some c_1 . Let, $P(c_3)$ be True for some $c_1 \neq c_3$. $\because c_2$ is arbitrary, $\therefore P(c_3) \rightarrow c=c_3$ is True. By Modus ponens, $(c=c_3)$ is True. This contradicts the assumption that $P(c)$ is True for some $c_1 \neq c_3$.

$\therefore c$ is the only unique element for which $\exists x (P(x) \wedge \forall y (P(y) \rightarrow x=y))$ is True. $\square$

18. Show that if a and c are real numbers and $a \neq 0$, then there is a unique solution of the equation $ax+b=c$.

Solution: **Existence:** $ax+b=c \Rightarrow ax=(c-b) \Rightarrow x = \frac{(c-b)}{a}$

Substituting x in $ax+b$, we get: $a(\frac{c-b}{a})+b = (c-b)+b = c+(-b+b) = c+0 = c$

$\therefore x = \frac{(c-b)}{a}$ is a solution of the equation $ax+b=c$.

Uniqueness: Let, there exist two integers x_1 and x_2 such that $(ax_1+b) = (ax_2+b)$

$(ax_1+b) = c \quad \therefore (ax_1+b)+(-b) = (ax_2+b)+(-b) \Rightarrow ax_1+(b+(-b)) = ax_2+(b+(-b))$

$\Rightarrow ax_1+0 = ax_2+0 \Rightarrow ax_1 = ax_2 \Rightarrow ax_1+(-ax_2) = ax_2+(-ax_2)$

$\Rightarrow a(x_1-x_2) = 0 \quad \because a \neq 0 \quad \therefore x_1-x_2 = 0 \Rightarrow x_1 = x_2 \quad \square$

19. Suppose that a and b are odd integers with $a \neq b$. Show there is a unique integer c such that $|a-c| = |b-c|$.

Solution: By Defn. 1, $\exists$ integers K_1 and K_2 s.t. $a = 2K_1+1$ and $b = 2K_2+1$.

$$ab_mean = ((2K_1+1) + (2K_2+1))/2 = K_1 + K_2 + 1$$

$$a - ab_mean = (K_1 - K_2) \quad b - ab_mean = (K_2 - K_1)$$